

Automatic Classification of Uveal Melanoma Regression Patterns Following Ruthenium-106 Plaque Brachytherapy Using Ultrasound Images and Deep Convolutional Neural Network

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Abstract

Following uveal melanoma (UM) affected treatment using ruthenium-106 brachytherapy, tumor thickness patterns fall into one of four categories: decrease (regression), increase (recurrence), stop (stable), or other, which are assessed in follow-up A-mode and B-mode images. These patterns are critical indicators of the tumor's response to therapy. This study aims to apply deep learning (DL) models for predicting post-brachytherapy tumor regression patterns. A cohort of 192 patients participated in this study. B-Mode images taken at the time of diagnosis were collected, and the ophthalmologists labeled the images into four regression patterns based on the results of the treatment. DenseNet121 and ResNet34 models were trained and evaluated using performance metrics. DenseNet121 achieved a macro-average AUC of 0.933, compared to 0.916 for the ResNet34. The per-class evaluation showed that DenseNet121 excelled in predicting all categories, providing superior predictive accuracy. The ablation study revealed that the best performance was achieved without pretrained weights, using dropout layers and a batch size of 32. Both models demonstrated strong classification capabilities, with DenseNet121 providing the highest overall accuracy. This study highlights the potential of DL models in predicting regression patterns in UM patients undergoing brachytherapy. Further validation and exploration of their integration into clinical practice are warranted.

Introduction

Uveal melanoma (UM) is the most prevalent and aggressive intraocular cancer, primarily affecting adults and frequently arising from the uveal tissue. Although new treatment methods such as proton therapy and gamma knife radiosurgery have emerged ¹, plaque brachytherapy, specifically with ruthenium-106 and iodine-125, remains the practical choice and preserves the globe. Numerous well-established factors, such as tumor size and thickness, location, retinal detachment, extrascleral extension, and retinal invasion, have been shown to influence anatomical outcomes ^{2,3}.

Tumor-specific genetic alterations and histopathologic characteristics, including epithelioid cell type, monosomy 3, 6p gain, and loss of the BAP-1 gene, are the key factors in predicting melanoma-specific patient outcomes ⁴. Although fine-needle aspiration (FNA) can provide gene alteration results, it is often not available in any center. ⁵ Ultrasound (US) imaging, on the other hand, is widely available and essential for assessing the dimensions of UM and monitoring tumor status evaluation during follow-up ⁵. Tumor regression patterns, including thickness and basal diameter, are routinely evaluated from initial diagnosis to follow-up sessions using B-scan ultrasonography, and subretinal fluid resorption is assessed by funduscopy ⁶.

Radiotherapy initiates tumor regression that is routinely evaluated in clinical practice as the change in tumor thickness measured by B-scan ultrasonography, and subretinal fluid resorption assessed by funduscopy ⁷. The regression pattern is a critical indicator of the response to plaque therapy treatment. A few reports have highlighted instances of local treatment failure ⁷⁻¹⁰, and subsequent enucleation, and others have indicated that rapid tumor regression following plaque brachytherapy ^{11,12} is often associated with a poor survival prognosis. Computer vision offers promising solutions to these challenges by assisting ophthalmologists in decision-making ^{13,14}, and existing models already incorporate clinical and demographic factors to predict individual patient prognosis following UM treatment ¹⁵⁻²¹.

Recently, artificial intelligence (AI) has shown significant progress in big data retrieval, precise feature extraction, and improved consistency and efficiency in medical image analysis ²²⁻²⁶. Deep learning (DL), a subset of AI, has become one of the most widely used approaches for medical image analysis, particularly in predicting cancers at early stages through convolutional neural networks (CNNs). This study represents an attempt to develop a DL-based predictive model for patients with UM following ruthenium-106 plaque brachytherapy, utilizing first and follow-up US images and clinical data. By investigating regression pattern prediction through DL, this work aims to establish a clinically accessible model to enhance prognosis and guide treatment decisions, addressing a critical gap in personalized post-brachytherapy monitoring.

Methods

Study workflow

Figure 1 illustrates the complete workflow adopted in this study, encompassing clinical procedures, data collection, and DL implementation. The objective is to explore the feasibility and effectiveness of DL in predicting UM regression patterns. The process begins with the identification of UM in patients (Step 1), followed by a clinical examination by an Ocular oncologist (Step 2). Ultrasound imaging is performed (Step 3), and plaque brachytherapy is administered by a multidisciplinary team of physicians and medical physicists (Step 4). B-scan ultrasound images are then collected (Step 5), along with clinical, demographic, dosimetric, and follow-up data (Step 6). These data are preprocessed through augmentation, resizing, normalization, and dataset splitting (Step 7), after which the DL model is developed (Step 8) and evaluated using standard performance metrics (Step 9). The final outcome is a multi-class prediction of UM regression patterns, including increase, decrease, stop, and other categories (Step 10).

Study cohort

This retrospective, single-center study was conducted at the Eye Research Center. Due to the retrospective nature of the study, the Ethics Committee of IUMS (IR.IUMS.FMD. REC.1402.417) waived the need of obtaining informed consent, and approved the experiments. All methods were performed in accordance with the relevant guidelines and regulations. The study population included adult patients clinically diagnosed with UM between July 2017 and 2022 who underwent treatment with ruthenium-106 plaque brachytherapy. The exclusion criteria included: (1) age under 18, (2) prior treatment with therapies other than ruthenium-106 plaque brachytherapy, (3) follow-up duration of less than two years, and (4) presence of metastatic disease at diagnosis.

B-Mode ultrasound images of 192 patients were collected at the time of uveal melanoma diagnosis and before the start of brachytherapy treatment. Each patient had at least two images in both transverse and longitudinal views at the time of diagnosis, so a total of 661 images were collected. Each patient had at least 2 years of follow-up, so the ocular oncologist labeled the patients' images into four groups in terms of regression pattern, increase, decrease, stop, and other, based on the response to treatment of the mass during follow-up sessions. Additionally, US images were independently interpreted and reported by two board-certified ophthalmologists specializing in ocular oncology, each with a minimum of 10 years of experience

Regression Patterns in Intraocular Melanoma Post-Brachytherapy

Patients were treated with ruthenium 106 (BEBIG GmbH, Berlin, Germany) plaques to deliver the appropriate dose to the apex and base of the tumor. Two plaque types were utilized: round (CCA, CCB, and CGD), COB (notch), and CIA. An experienced medical physicist oversaw the dosimetry process, ensuring accurate dose delivery based on the activity and half-life of the ruthenium-106 plaques. The physicist also documented the dosimetry reports for each treatment. An ocular oncologist measured tumor thickness and the largest basal diameter (LBD). Tumor thickness was assessed from the inner scleral surface to the tumor apex along two meridians: one along the LBD and another perpendicular to it. Representative digitized scans were prospectively stored during each diagnostic and follow-up visit. We tracked the changes in tumor thickness and the LBD over time and classified tumor regression patterns according to Abramson et al.²⁷, using four main categories: D (decrease; a progressive reduction in thickness by at least 15% post-brachytherapy), S (stable; less than 15% change in thickness), I (increase; a progressive increase in thickness by at least 15%), and others²⁷. The "others" category was further divided into five subtypes: DS (decrease followed by stability), DI (decrease followed by increase), ID (increase followed by decrease), SD (stability followed by decrease), and zigzag (alternating measurements with no clear trend). A total of 661 images were collected from 192 patients with intraocular melanoma before the beginning of treatment. All images were acquired by a single operator and independently interpreted by two ophthalmologists, each with over ten years of experience in plaque radiotherapy for UM. As illustrated in Fig. 2, B-Mode US images of four different patients with Decrease, Other, Stop, and Increase regression pattern at the time of diagnosis (A1, A2, A3, and A4), one year (B1, B2, B3, and B4), and two years (C1, C2, C3, and C4) post-treatment treatment with Ru-106 plaque brachytherapy.

Data preprocessing

All US images were obtained in JPG format from the US imaging databases. The dataset was processed using Python version 3.12.4 and libraries, including NumPy version 1.26.4, PIL version 10.3.0, Albumentations version 2.0.5, and Torchvision version 0.21.0 + cu124. A 5-fold cross-validation (CV) approach was applied to split the dataset, using 80% for training and 20% for testing in each fold. To address the class imbalance, oversampling was performed by applying horizontal flipping, rotation, shifting, scaling, and brightness/contrast adjustments. This process ensured that all classes contained equal number of images.

Further data augmentation was conducted to enhance variability. Each image was transformed, including random gamma correction, slight rotation, Gaussian blur, and brightness/contrast modifications, generating five augmented versions per image. This step increased dataset diversity and mitigated overfitting. The number of images was 661, and after augmentation increased to 5493 images. Before model training, images were resized to 224×224 pixels to match input requirements. Normalization was applied using a mean of [0.485, 0.456, 0.406] and a standard deviation of [0.229, 0.224, 0.225] and conversion to tensor format, ensuring consistency with pre-trained models.

Deep Learning Model Construction

Two convolutional neural network architectures—DenseNet121 and ResNet34—were utilized as the backbone models for image multi-class classification. These architectures were chosen due to their proven ability to perform well in image classification tasks while maintaining reasonable computational efficiency. Both models were initialized with random weights (weights = None) and trained under identical conditions to ensure a fair comparison.

DenseNet-121 is composed of five dense blocks organized across three transition levels, with each block containing multiple convolutional layers followed by a max-pooling operation. The architecture leverages dense connectivity, where each layer receives input from all preceding layers, promoting feature reuse and mitigating the vanishing gradient problem. The final output, after global average pooling, is passed through a series of fully connected layers, followed by a dropout layer ($p = 0.4$) tailored for four-class classification.

ResNet34, on the other hand, is structured around residual learning, consisting of an initial convolutional layer followed by four stages of residual blocks, each incorporating identity shortcut connections. These skip connections allow gradients to propagate more effectively during training, enabling deeper network architectures without degradation in performance. In the case of ResNet34, the final fully connected layer was similarly modified to include a linear transformation and a dropout layer with the same probability of performing four-class classification. Both models were trained using 2D ultrasound images and the corresponding ground truth labels provided in the dataset. Input images were resized to 224×224 pixels and normalized using ImageNet statistics.

Training was performed using stochastic gradient descent (SGD) with a learning rate of 0.001, momentum of 0.9, and weight decay of $1e-4$. The loss function used was cross-entropy with label smoothing ($\epsilon = 0.1$). A five-fold CV approach was implemented to ensure robustness. Training was conducted over 200 epochs, where each epoch involved forward propagation, loss computation, gradient backpropagation, and weight updates.

Ablation Study

To evaluate the contribution of individual components within the model architecture and training strategy, a series of ablation experiments were performed. Each experiment involved modifying a specific element while keeping all other parameters constant to isolate its impact on model performance. Since

DenseNet121 outperformed ResNet34, all six ablation scenarios were conducted exclusively for DenseNet121: (1) initializing the network with pre-trained ImageNet weights, (2) reducing the batch size to 16, and (3) removing dropout layers from the architecture. The results of these experiments provided insights into the sensitivity of the model to various training configurations and optimization strategies.

Performance evaluation

The model performance is validated on the test set after each epoch, with key performance metrics logged for analysis. These metrics provide a comprehensive assessment of the model's performance:

- Confusion Matrix, Accuracy, Recall (Sensitivity), Precision, F1 Score, ROC Curve, and AUC (area under the curve).
 - Macro average computes the AUC for each class individually and then averages the results. This approach treats all classes equally, regardless of their frequency in the dataset.
 - Micro average aggregates the contributions of all classes to compute a global AUC score.

These averages provide a broader view of model performance across all classes.

- Cohen's Kappa Coefficient: The Kappa coefficient evaluates agreement between predicted and actual classifications while accounting for chance agreement. It ranges from - 1 to 1, where a Kappa of 1 indicates perfect agreement, 0 suggests no agreement beyond chance, and values below 0 reflect worse-than-random performance, indicating poor model reliability. P-values < 0.05 were considered statistically significant.

These metrics together offer a robust evaluation of model performance across multiple dimensions.

Results

Baseline Characteristics

A total of 192 patients with UM and treated with plaque brachytherapy were included in this study. The cohort comprised 60.5% females and 39.5% males, with choroidal melanoma being the most common type (63%) and ciliary body involvement in 37%. Tumor sizes were evenly distributed between small (26%), medium (26%), and large (35.5%), with 12.5% classified as very large. Most of the tumors were dome-shaped (74%) and located in the temporal part (28%). The most common inserted plaques were CGD (34.9%) and CCB (28.1%). Enucleation and recurrence rates were 3.6% and 19.8%, respectively. Also, 85 (44.3%) of patients underwent Transpupillary Thermotherapy (TTT), 67 (34.9%) of which were performed as adjuvant therapy, and 18 (9.4%) were prescribed due to tumor recurrence, however, in 10 of them, TTT was not effective enough as a secondary treatment for recurrence, so replaque therapy was done. The right eye was affected in 52.1% of cases. Diabetes was present in 17.2% of patients. The mean age was 52.5 ± 14.1 years, and the apex and scleral doses averaged 85.5 ± 25.0 Gy and 906.1 ± 478.6 Gy, respectively, with a mean radiation time of 116.9 ± 92.3 hours. Also, the regression patterns after plaque radiotherapy showed 4.8%, 51.5%, 27.2%, and 16.5% increase, decrease, other, and stop, respectively.

Model Evaluation

To assess the overall classification performance of the two models, DenseNet121 and ResNet34, a five-fold CV approach was employed. The averaged training and validation metrics across folds are summarized in Table 1. This table provides a comprehensive view of each model's global performance, not per class, including accuracy, precision, recall, F1 score, and Cohen's Kappa. In each metric, the model with better performance is underlined. These metrics reflect how well the models generalize across different subsets of the data and serve as a benchmark for overall model robustness.

Table 1
Mean performance metrics across 5-fold cross-validation for ResNet34 and DenseNet121 models.

Metrics	Dataset	Resnet34	Densenet121
Accuracy	Train	0.81	0.83
	Validation	0.77	0.80
Precision	Train	0.85	0.84
	Validation	0.74	0.77
Recall	Train	0.81	0.82
	Validation	0.73	0.76
F1 Score	Train	0.81	0.81
	Validation	0.72	0.76
Kappa (p < 0.05)	Train	0.71	0.78
	Validation	0.60	0.63

To further understand model behavior across specific output categories, we conducted a per-class analysis. The performance metrics for each class—Decrease, Increase, Other, and Stop—are detailed separately for DenseNet121 and ResNet34 in Tables 2 and 3, respectively. These tables report the class-wise training and validation accuracy, precision, recall, and F1 scores, based on the epoch where each class achieved its highest validation accuracy. This per-class breakdown provides insights into model strengths and potential limitations when distinguishing among specific therapeutic response categories.

Table 2
Per-class performance metrics for the DenseNet121 model.

Category	Train Accuracy	Val Accuracy	Train Precision	Val Precision	Train Recall	Val Recall	Train F1 Score	Val F1 Score
Decrease	0.88	0.87	0.87	0.83	0.92	0.87	0.87	0.85
Increase	0.97	0.91	0.90	0.87	0.91	0.88	0.90	0.88
Other	0.92	0.88	0.92	0.86	0.86	0.81	0.88	0.82
Stop	0.93	0.92	0.95	0.93	0.86	0.84	0.89	0.88
Mean	0.92	0.89	0.91	0.87	0.89	0.85	0.88	0.86

Table 3
Per-class performance metrics for the Resnet34 model.

Category	Train Accuracy	Val Accuracy	Train Precision	Val Precision	Train Recall	Val Recall	Train F1 Score	Val F1 Score
Decrease	0.87	0.85	0.85	0.83	0.91	0.85	0.85	0.85
Increase	0.97	0.91	0.90	0.88	0.88	0.78	0.88	0.82
Other	0.92	0.85	0.92	0.81	0.85	0.81	0.88	0.80
Stop	0.90	0.91	0.96	0.89	0.89	0.84	0.88	0.87
Mean	0.915	0.88	0.90	0.85	0.88	0.82	0.87	0.83

For a more focused comparison of validation performance between the two models, we summarized the key metrics side by side in Table 4. This comparative table highlights the differences in classification ability between DenseNet121 and ResNet34 for each class on the validation set. It includes metrics such as validation accuracy, precision, recall, F1 score, and AUC, allowing a nuanced evaluation of which model performs better for each therapeutic response type. In each metric, the model with better performance is underlined>. This analysis is particularly valuable for identifying model-specific strengths and for guiding the choice of architecture in clinical decision support contexts.

Table 4
Comparison of DenseNet121 and ResNet34 models based on mean validation set performance for each response class.

Class/ Model	Accuracy		Precision		Recall		F1 Score		AUC	
	Resenet34	Densenet121	Resenet34	Densenet121	Resenet34	Densenet121	Resenet34	Densenet121	Resenet34	Densenet121
Decrease	0.85	0.87	0.83	0.83	0.85	0.87	0.85	0.85	0.89	0.90
Increase	0.91	0.91	0.88	0.87	0.78	0.88	0.82	0.88	0.91	0.90
Other	0.85	0.88	0.81	0.86	0.81	0.81	0.80	0.82	0.87	0.88
Stop	0.91	0.92	0.89	0.93	0.84	0.84	0.87	0.88	0.93	0.95
Mean	0.88	0.89	0.85	0.87	0.82	0.85	0.83	0.86	0.89	0.90

Figure 3 illustrates the classification performance of the two evaluated models—ResNet34 and DenseNet121—on the validation set from fold 0. Panel (a) shows that ResNet34 achieved strong class separation, particularly for the 'Stop' and 'Other' classes, although some confusion remained between 'Decrease' and 'Other'. The corresponding ROC curves reveal high AUC values across all classes, with a macro-average AUC of 0.916. In comparison, panel (b) demonstrates that DenseNet121 achieved superior performance, correctly classifying a larger number of 'Decrease' and 'Increase' samples, as reflected in its confusion matrix. The ROC analysis further supports this observation, with DenseNet121 reaching higher AUCs, including 0.984 for 'Increase' and a macro-average AUC of 0.933.

Ablation Study

Table 5 provides detailed validation metrics, including accuracy, precision, recall, F1-score, and AUC for each scenario across all classes. The final column corresponds to the baseline configuration, included for comparison with the other three scenarios, and it is evident that it achieves the best overall performance among all. Additionally, Fig. 4 (a to c) presents the confusion matrices and ROC curves for each ablation scenario (1–3), corresponding to

subfigures (a) through (c), respectively. The results demonstrate that the baseline configuration—employing dropout layers, no pretrained weights, and a batch size of 32—outperformed all other ablation variants. Specifically, deviations from this setup, such as removing dropout, incorporating pretrained weights, or reducing batch size, resulted in reduced classification performance. These findings highlight the robustness and effectiveness of the main training configuration for DenseNet121 in this context.

Table 5
Validation metrics for the three ablation scenarios across all classes using the DenseNet-121 model. The final column corresponds to the baseline configuration (with dropout, no pretrained weights, batch size of 32), which achieved the best overall performance.

Class	Metrics\Scenarios	1: With pretrained ImageNet weights	2: With a batch size of 16	3: Without dropout	Main
Decrease	Accuracy	0.82	0.76	0.78	0.87
	Precision	0.83	0.78	0.79	0.85
	Recall	0.82	0.76	0.78	0.87
	F1 Score	0.80	0.76	0.78	0.85
	AUC	0.88	0.84	0.83	0.90
Increase	Accuracy	0.97	0.97	0.95	0.91
	Precision	0.95	0.98	0.70	0.98
	Recall	0.76	0.71	0.59	0.88
	F1 Score	0.80	0.79	0.61	0.88
	AUC	0.92	0.89	0.98	0.90
Other	Accuracy	0.86	0.82	0.84	0.88
	Precision	0.84	0.77	0.82	0.75
	Recall	0.78	0.73	0.74	0.81
	F1 Score	0.80	0.75	0.76	0.82
	AUC	0.88	0.84	0.84	0.88
Stop	Accuracy	0.90	0.91	0.89	0.92
	Precision	0.86	0.89	0.86	0.87
	Recall	0.80	0.82	0.78	0.84
	F1 Score	0.83	0.85	0.80	0.88
	AUC	0.92	0.91	0.92	0.95

Discussion

Recent advancements in AI have significantly transformed medicine, particularly in the analysis of large datasets, enabling more accurate diagnostic and prognostic predictions. Numerous studies have highlighted the potential of computer-aided systems in enhancing clinicians' diagnostic capabilities^{28,29}. Understanding prognosis, both pre- and post-surgery, is crucial for patients, reducing anxiety and enabling future planning³⁰⁻³². Deep learning models offer efficient and accurate predictions and classifications for diverse diseases using imaging data, including breast cancer²⁸, liver diseases²⁹, colon cancer³⁰, brain tumor³¹, skin cancer³², lung cancer³³, and pneumonia³⁴, and COVID-19³⁴, all with minimal human oversight.

In terms of classifying, diagnosing, and prognosing UM, previous studies have applied a combination of different imaging modalities, including US, fundus photography, and optical coherence tomography (OCT), along with AI, to improve accuracy and predictive performance³⁵. Notably, this study represents the first attempt to use deep learning models to predict the regression pattern of UM using pre-treatment ultrasound images, which are labeled by ocular oncologists based on the results of treatments. By addressing this gap, our work provides a novel approach to personalizing treatment plans for UM patients and enhancing the predictive power of clinical decision-making.

Previous AI-based studies in UM have leveraged different imaging techniques to classify and diagnose ocular melanoma. For example, Biswarup et al.³⁶ developed a CNN-based method for detecting ocular melanoma using fundus images from 170 pre-melanoma patients, while Zhang et al.³⁷ explored the correlation between iris color and UM. AI, combined with image modalities, can also differentiate UM from nevus and other lesions. In this regard, Ahmed et al.³⁸ employed the US images and an artificial neural network (ANN) approach for intraocular melanoma classification. Yao et al.³⁹ applied DensNET21 and FP to classify uveal melanoma and choroidal nevi, and examined the color fusion effect on UM and uveal nevus classification, and Dadzie et al.⁴⁰ applied DenseNet121 and FP to an automated diagnosis of UM. Goswami et al.⁴¹ evaluated DL models for distinguishing benign from malignant intraocular tumors, and finally, Sabazade et al.⁴² developed a DL model to differentiate small choroidal melanomas from nevi, while Hoffmann et al.⁴³ evaluated DL for

classifying choroidal melanocytic lesions based on FP images. Valmaggia et al.⁴⁴ assessed OCT-based auto-segmentation of pigmented choroidal lesions (PCLs) using DL. Additionally, Jiechao Ma et al.⁴⁵ employed ultra-wide-field fundus images and DeepLabv3 architecture for UM segmentation, presenting a model with high sensitivity for the early diagnosis of UM tumors. Despite these advancements, no study has yet attempted to predict UM's regression pattern using pre-treatment ultrasound images—an area this study pioneers, ultimately contributing to more accurate prognosis and tailored therapeutic strategies for patients.

This study specifically characterizes the post-brachytherapy regression patterns of UM by analyzing initial tumor thickness and LBD, which emerged as powerful and valuable predictors of treatment outcomes. Abramson et al.²⁷ analyzed 82 UM patients to assess tumor regression patterns, reporting that no two UMs regressed identically after brachytherapy. In their study, of cases exhibited progressive regression (pattern D), 16% remained stable (pattern S), 12% increased in size (pattern I), and 2% followed other regression patterns (pattern O). In contrast, our study identified 4.8% of UM increased over time, with 51.5% following pattern D, 27.2% pattern O, and 16.5% pattern S. Rui Fang et al.⁴⁶ found that 36% of UMs remained stable, while 46.8% exhibited pattern D regression, similar to the results of Rashid et al.⁴⁷. A possible explanation for the differences observed could lie in the use of different isotopes, as the type of isotope has been reported as an independent predictor of tumor regression⁴⁸. Additionally, the tumor's original height was positively associated with regression rate, aligning with the findings of Rashid et al.⁴⁸, which indicated that larger tumors tend to shrink significantly faster than smaller ones.

In this study, two CNN architectures—DenseNet-121 and ResNet-34—were utilized as backbone models for multi-class image classification of regression patterns in UM-affected patients, based on their pretreatment US images. These models were selected after evaluating several architectures, including deeper versions of ResNet and DenseNet, as well as VGG-based networks. While the alternative architectures exhibited signs of overfitting or instability (e.g., larger test-validation performance gaps and fluctuating validation curves), DenseNet-121 and ResNet-34 demonstrated more stable learning behavior and consistent performance.

The performance of both the ResNet34 and DenseNet121 models was evaluated using a five-fold CV approach, with results summarized in Table 1. Across all metrics, DenseNet121 demonstrated a slight edge over ResNet34, particularly in validation accuracy, precision, and recall. Specifically, DenseNet121 achieved a validation accuracy of 80%, compared to 77% for ResNet34, reflecting its ability to generalize slightly better to unseen data. Although both models exhibited similar performance on training data, DenseNet121 maintained a consistent advantage in precision (77% vs. 74%) and recall (76% vs. 73%), indicating its slightly better capability to balance false positives and false negatives. The F1 score, which combines precision and recall, also favored DenseNet121 with a validation score of 0.76, compared to 0.72 for ResNet34. Cohen's Kappa, a measure of agreement beyond chance, revealed that DenseNet121 outperformed ResNet34 in training (0.78 vs. 0.71), though the difference in validation was smaller (0.63 vs. 0.60). These results suggest that DenseNet121 is a slightly more robust model overall, although both models perform well and exhibit similar trends in their ability to generalize across different data subsets.

The per-class analysis in Tables 2 and 3 highlights key differences in the performance of the DenseNet121 and ResNet34 models across therapeutic response categories. DenseNet121 generally showed strong performance, particularly in recall for the "Decrease" and "Stop" categories, achieving the highest validation accuracy of 0.87 in the "Decrease" class. However, it slightly underperformed in recall for the "Increase" class (0.88). ResNet34, while similar, showed slightly lower recall in the "Increase" and "Other" categories (0.78 and 0.81), but excelled in precision for the "Stop" category (0.93).

In Table 4, DenseNet121 slightly outperforms ResNet34 in validation accuracy (0.89 vs. 0.88) and F1 score, particularly in the "Decrease" and "Stop" categories. ResNet34 performed better in recall for the "Increase" class (0.88 vs. 0.78) and precision for "Stop" (0.93 vs. 0.89). Overall, DenseNet121 offers more balanced performance across categories, while ResNet34 excels in identifying "Stop" responses with high precision. This analysis is crucial for choosing the appropriate model for specific therapeutic response classifications, with DenseNet121 being more consistent and ResNet34 better suited for precision-focused tasks.

Additionally, as shown in Fig. 3, DenseNet121 consistently outperformed ResNet34 in terms of classification accuracy and discriminative ability across all classes. The improved performance is evident both in the confusion matrix, where DenseNet121 shows fewer misclassifications for the 'Decrease' and 'Increase' categories, and in the ROC analysis, which reports higher AUC values for each class. This suggests that DenseNet121 may have better capacity to learn feature representations relevant to our classification task. Notably, while ResNet34 also performed well (macro-average AUC = 0.916), it exhibited greater confusion between 'Decrease' and 'Other', which could impact its reliability in a real-world setting where precise class separation is crucial. These findings highlight the robustness of DenseNet121, especially in distinguishing subtle inter-class variations.

The results in Table 5 and Fig. 3 demonstrate that the baseline configuration—using dropout layers, no pretrained weights, and a batch size of 32—consistently outperformed the other ablation scenarios across all evaluation metrics. Deviations from this setup, such as removing dropout, incorporating pretrained weights, or reducing batch size, led to reduced performance, particularly in recall and F1-score for the "Increase" class. This highlights the robustness and effectiveness of this configuration in the given classification task.

Using pretrained ImageNet weights (Scenario 1) resulted in a decrease in overall performance for most classes, particularly in the "Increase" class, where accuracy dropped from 0.97 to 0.91. This suggests that pretrained weights may not have been as beneficial for this particular task, potentially due to domain differences between the source dataset (ImageNet) and the current task's data. Similarly, reducing the batch size (Scenario 2) to 16 and removing dropout (Scenario 3) also led to performance declines across all metrics. For instance, in Scenario 2, the "Increase" class saw a drop in recall (0.76 to 0.71) and precision (0.83 to 0.79), while Scenario 3 showed a notable decrease in AUC and precision for the "Decrease" class. These results indicate that both batch size and dropout play critical roles in stabilizing the model's performance, with dropout particularly contributing to improved generalization.

Figure 3 (a to c) presents confusion matrices and ROC curves for each ablation scenario, further supporting these conclusions. Additionally, we experimented with Adam and AdamW optimizers, but both led to severe overfitting and were therefore excluded from the reported results. The primary densenet121 model employed SGD with a learning rate of 0.001, momentum of 0.9, and weight decay of 1e-4, yielding the most stable and generalizable performance across all classes.

This work has several limitations, including the use of a relatively small dataset, data acquired from a single center, and the absence of external validation. The limited dataset size restricts the model’s ability to generalize across broader populations and imaging settings. Data scarcity is a common challenge in medical imaging, particularly for rare conditions such as choroidal melanoma, and is further exacerbated by privacy concerns and limited access to annotated datasets. Collaborative multicenter studies and the development of shared databases are critical next steps to overcome these limitations and support model robustness.

Furthermore, this study used 2D JPEG ultrasound images, which lack the full spatial and intensity information available in high-resolution 3D imaging. Future research should explore training with full-resolution or 3D ultrasound data, enabling the capture of richer spatial features and potentially improving diagnostic accuracy.

Beyond CNN-based models, future research could investigate the use of vision transformers, which have shown promise in capturing global context and improving performance in various medical imaging tasks. Moreover, the incorporation of explainable AI (XAI) methods, such as saliency maps, Grad-CAM, or concept-based attribution, would provide transparency into model decisions and foster trust among clinicians. Combining imaging data with clinical variables or biomarkers through multimodal learning approaches could also enhance model performance and lead to more personalized predictions.

Conclusions

This study marks an attempt to develop a CNN-based model, using DenseNet121, to predict regression patterns of UM following brachytherapy based on pre-treatment US images. The model successfully classified responses into four distinct categories—Increase, Decrease, Stop, and Other—achieving the highest accuracy in predicting the patterns. DenseNet121 demonstrated balanced performance across all classes, highlighting its potential as a tool for early treatment response assessment and personalized treatment planning. As the first of its kind, this approach lays the foundation for future advancements, where larger, multi-center datasets could further enhance model performance, enabling more robust and generalized predictions.

Abbreviations

- AI**
artificial intelligence
- ANN**
artificial neural networks
- AUC**
area under the curve
- CNN**
convolutional neural network
- CV**
cross validation
- DL**
deep learning
- DNS**
dysplastic nevus syndrome
- FC**
fully connected
- FN**
False negative
- FNAB**
fine needle aspiration biopsy
- FP**
fundus photography
- FP**
False Positive
- LBD**
largest basal diameter
- ML**
machine learning
- MRI**
magnetic resonance images
- NCI**
national cancer institute

OCT

Optical Coherent Tomography

PCLs

pigmented choroidal lesions

PCA

principal component analysis

ROC

receiver operating characteristic

SD

standard deviation

SGD

stochastic gradient descent

TN

True negative

TP

True positive

TTT

transpupillary thermotherapy

UM

uveal melanoma

US

ultrasound

Declarations

Data availability

The datasets used and analyzed during the current study will be available from the corresponding author on reasonable request.

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Competing interests

The authors declare no competing interests.

Ethics approval

The Ethics Committee of IUMS waived the requirement for ethical approval (IR.IUMS.FMD. REC.1402.417).

Author's contribution

Study conception and design: A.T, M.S, M.N; Study supervision and conceptual feedback: M.S, M.N; Data collection and preprocessing: A.T; Image interpretation and clinical oversight: M.N, R.M; Machine learning model development and analysis: E.Y, A.T; Result interpretation: A.T, M.N, R.M, M.S; Manuscript writing: A.T, E.Y. All authors read, edited, and approved the final manuscript.

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Figures

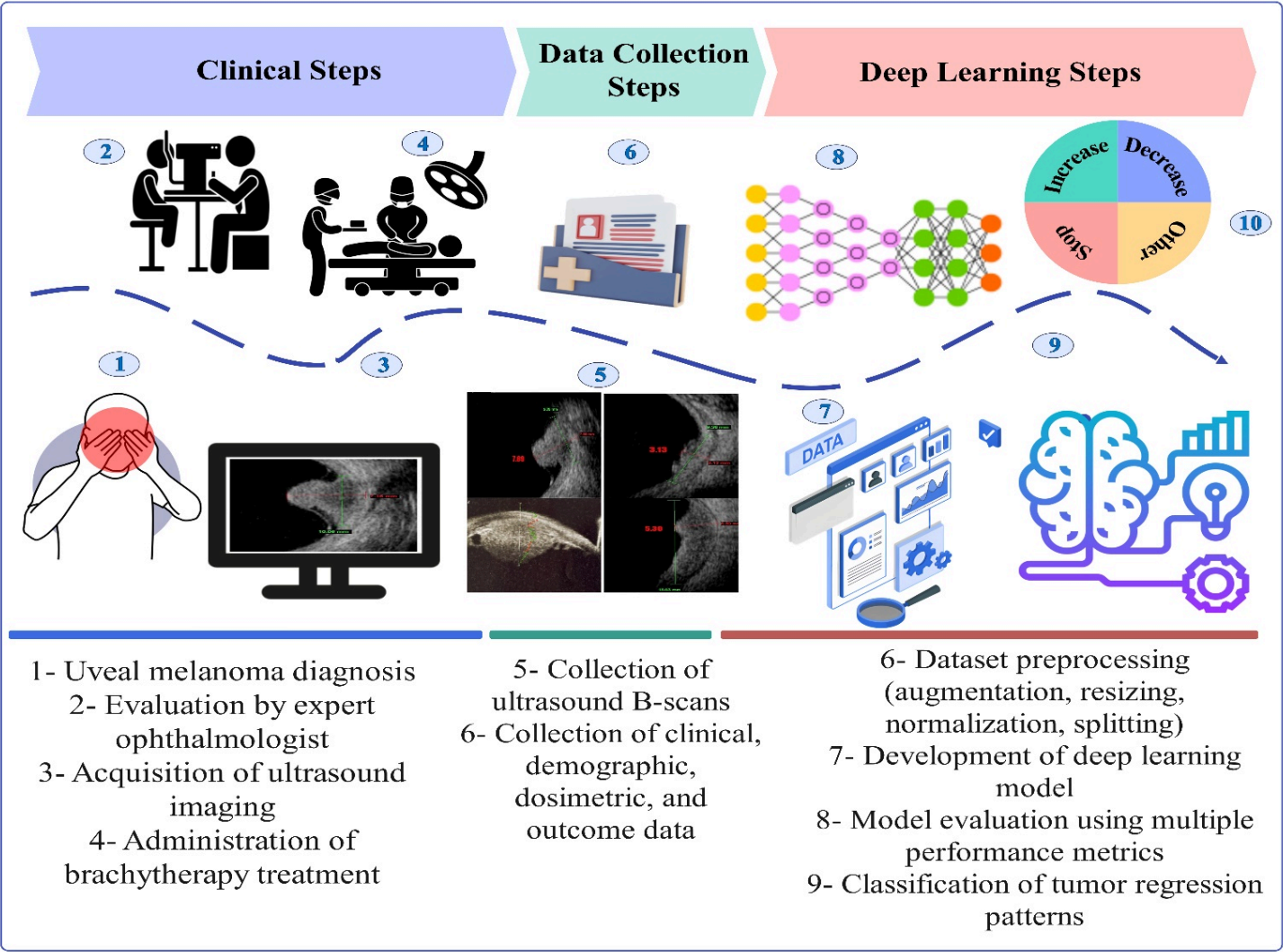


Figure 1

Workflow of the study illustrating the steps from UM diagnosis and plaque brachytherapy to deep learning model development. The process includes clinical examination, ultrasound imaging, data collection, preprocessing, and model evaluation, ultimately predicting UM regression patterns.

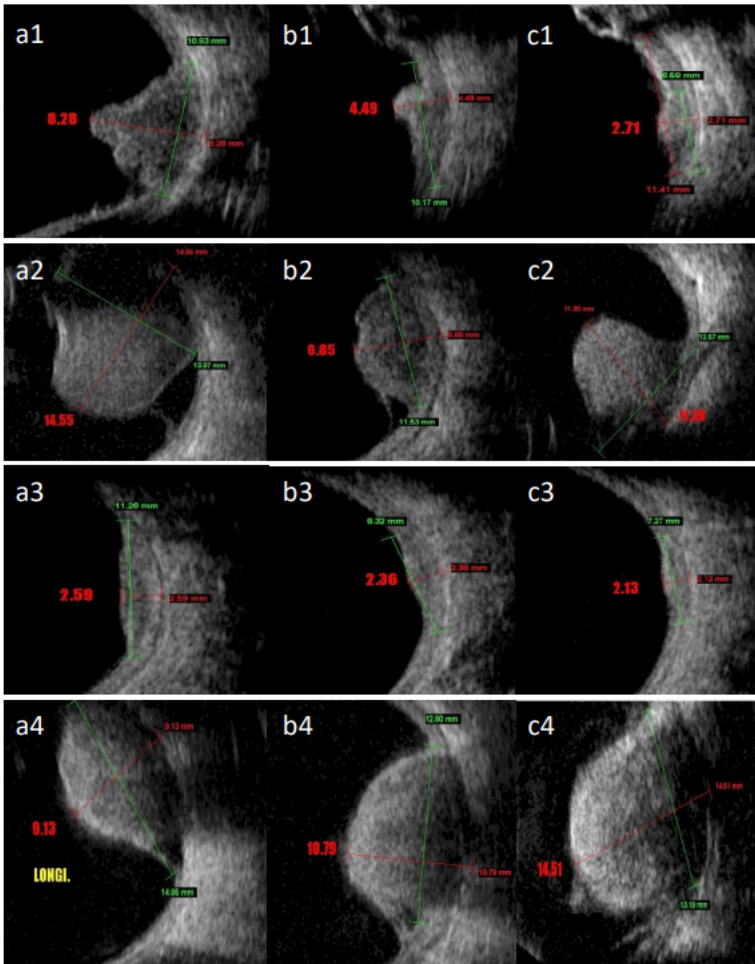
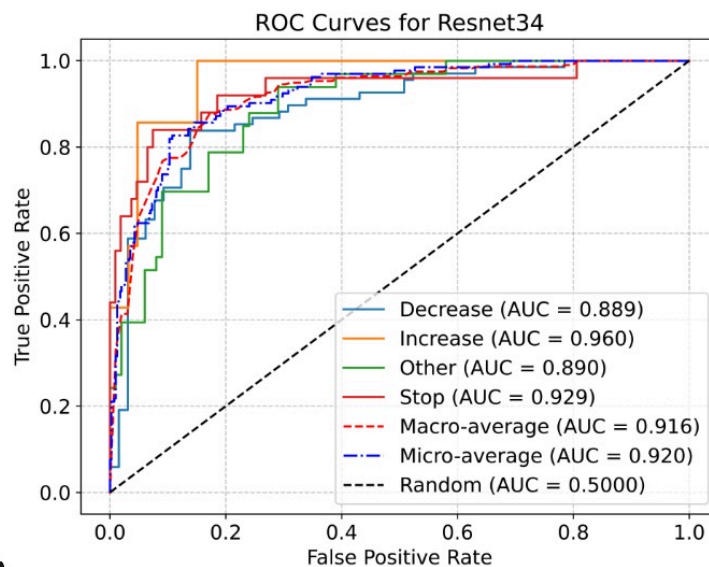
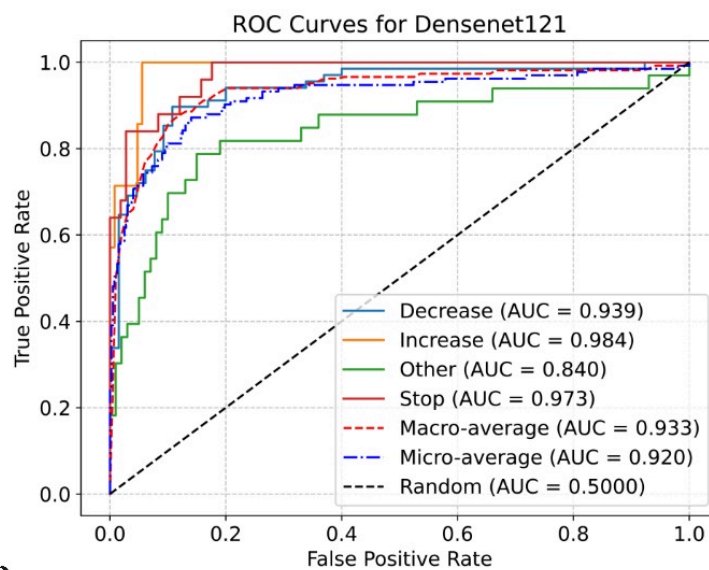
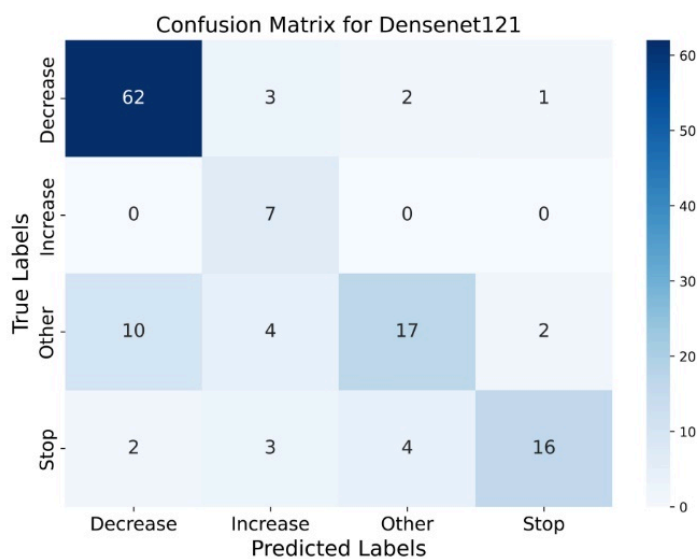


Figure 2

B-Mode ultrasound images of four different patients with Decrease, Other, Stop, and Increase regression pattern at the time of diagnosis (a1, a2, a3, and a4), one year (b1, b2, b3, and b4), and two years (c1, c2, c3, and c4) after treatment with Ru-106 plaque brachytherapy.



a



b

Figure 3

Model performance on the validation set from fold 0 (randomly selected). (a) ResNet34 and (b) DenseNet121, showing confusion matrices and ROC curves for each model.

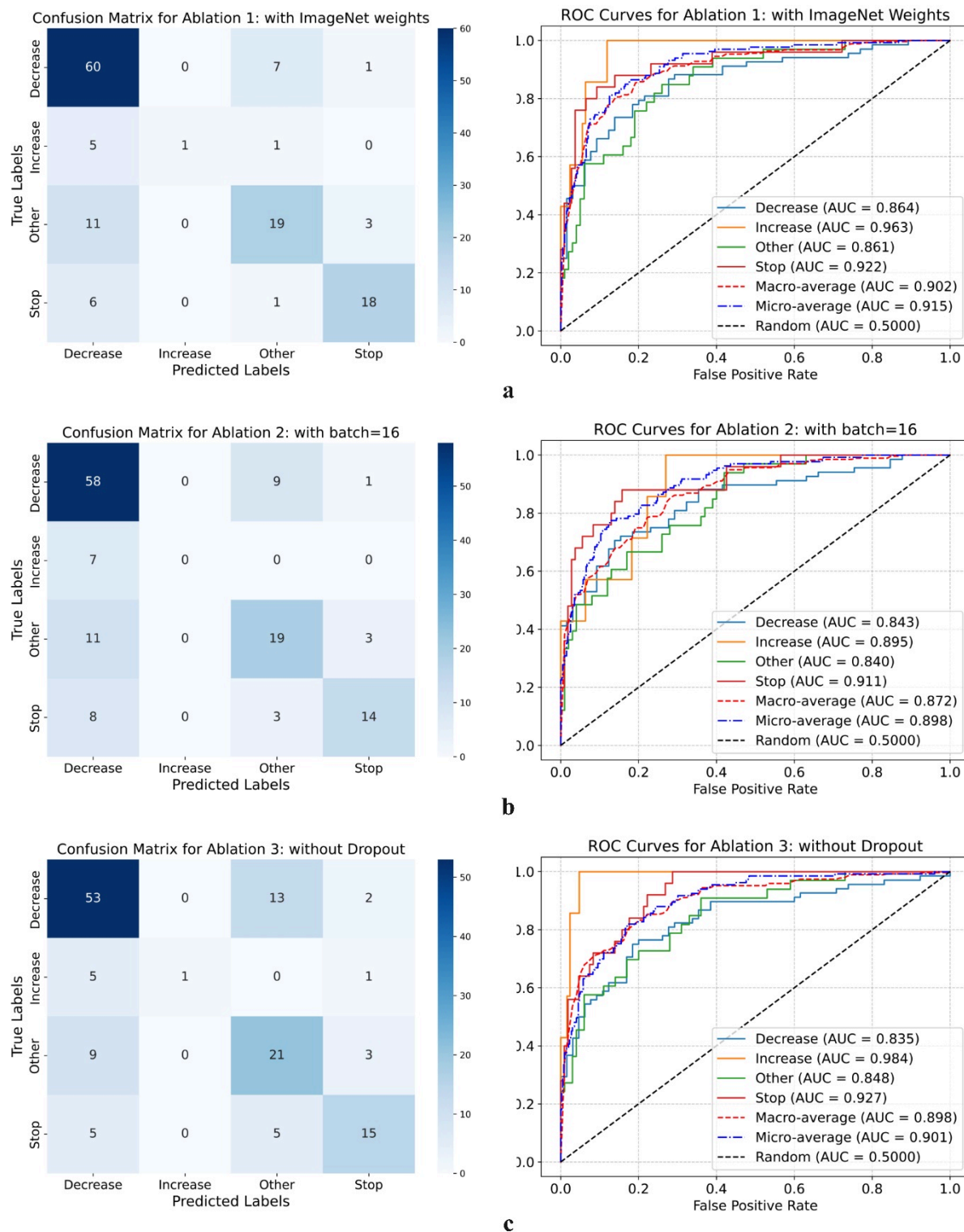


Figure 4

Confusion matrices and ROC curves from fold 0 (randomly selected) for the three ablation scenarios using the DenseNet-121 model: (a) with pretrained ImageNet weights; (b) reduced batch size (16); (c) without dropout.